AgriPower User Interface Handover

What this is

A web interface for running the AgriPower analysis and viewing the results. It is a front end only, it collects the inputs, runs the model in current_code/ and displays what the model produces.

What you need

- Windows.
- Python 3.11 (tested with 3.11.9).
- An internet connection on the first run.
- A Gurobi licence.

Install

Run these commands from the AgriPower folder, which is the folder that contains app.py:

python -m venv .venv

.venv\Scripts\python.exe -m pip install --upgrade pip

.venv\Scripts\python.exe -m pip install -r requirements.txt

Start the interface

From the AgriPower folder, run:

.venv\Scripts\python.exe -m streamlit run app.py

Then open http://localhost:8501 in a browser. If that port is busy, Streamlit uses the next free one and prints the address. Use the virtual environment's Python as shown, because a different Python can cause an import error.

Using the interface

1. Open New analysis. Upload the meter file, or select Use sample farm to try the app with the sample data.
2. Fill in the farm details if known, then select Save farm constraints. Any blank field uses the model's default.
3. Choose the scenario: farmer ownership or third party ownership.
4. Choose the year. The choice appears only if the file holds more than one complete year.
5. Select Start and note the 6 digit PIN.
6. Open Results, enter the PIN and select Check status. The page updates itself while the analysis runs.

Run time

A full year analysis takes about 9 minutes for farmer ownership and about 14 minutes for third party ownership. Only one analysis runs at a time. Jobs are lost if the app is restarted.

Outputs

- Figure: the model's four panel figure, which can be downloaded as a PNG.
- Results table: one column per option, shown on the page.
- PDF report: a two page summary of the recommended option and the farm inputs used.
- CSV: the results table, one row per option.